

# Bad-Character Sparsity and the High- $\ell$ Kummer Non-Concentration Barrier

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## Abstract

We record a structural bad-character criterion for prime Hecke non-concentration (PHNC) in the fixed-radical Kummer family attached to 2 and 3. A single near-trivial Kummer character forces its entire nontrivial cyclic orbit, of size  $\ell - 1$ , into a medium-deficit bad set. Consequently any analytic theorem bounding that bad set below  $\ell - 1$  proves PHNC. We then prove the exact finite mean-square identity for this bad set and show that the resulting route is not an unconditional high- $\ell$  proof: it is equivalent to deterministic vector-fiber equidistribution of the Kummer log vectors. This exposes a precise barrier and promotes the needed analytic input as a Kummer-sparsity hypothesis, not as a consequence of classical Bombieri–Friedlander–Iwaniec or generic large-sieve technology.

## 1 Purpose

The criterion paper isolates PHNC as the top analytic input in the Erdős–Graham #203 program. Paper 12 proves the Bernoulli-Kummer smoothing theorem once weighted PHNC is supplied. Paper 13 proves that an arbitrary-coefficient rank-two large-sieve bridge is false because finite Plancherel has a diagonal vector-fiber term.

This note attacks PHNC directly through bad-character counting. The outcome is a clean structural criterion and a clean barrier. It is not an unconditional proof of the high- $\ell$  seam.

## 2 PHNC and Kummer characters

Let  $\ell$  be prime,  $Q \geq 3$ , and

$$\mathcal{Q}_\ell(Q) = \{q \leq Q : q \equiv 1 \pmod{\ell}, q \nmid 6\}.$$

For  $q \in \mathcal{Q}_\ell(Q)$ , choose a primitive root  $g_q \pmod{q}$  and write

$$2 = g_q^{u_q}, \quad 3 = g_q^{v_q}.$$

Put

$$x_q = (u_q, v_q) \in \mathbb{F}_\ell^2.$$

For  $A = (r, s) \in \mathbb{F}_\ell^2 \setminus \{0\}$ , define

$$\chi_A(q) = e^{2\pi i A \cdot x_q / \ell}.$$

The condition  $A \cdot x_q = 0$  is equivalent to

$$2^r 3^s \in (\mathbb{F}_q^*)^\ell.$$

**Definition 1** (Prime average and deficit). Let  $N = |\mathcal{Q}_\ell(Q)|$ . For  $A \neq 0$ , define

$$M_Q(A) = \frac{1}{N} \sum_{q \in \mathcal{Q}_\ell(Q)} \chi_A(q), \quad \delta_Q(A) = 1 - |M_Q(A)|.$$

For  $0 < \eta < 1$ , define

$$B_\eta(\ell, Q) = \{A \in \mathbb{F}_\ell^2 \setminus \{0\} : \delta_Q(A) \leq \eta\}.$$

### 3 Power-orbit amplification

**Lemma 1** (Power-stability). Let  $f : X \rightarrow S^1$  on a finite probability space and put

$$\delta(f) = 1 - |\mathbb{E}f|.$$

Then for every integer  $m \geq 1$ ,

$$\delta(f^m) \leq m^2 \delta(f).$$

*Proof.* Choose  $\omega \in S^1$  so that  $\omega^{-1} \mathbb{E}f = |\mathbb{E}f|$ , and write  $\omega^{-1}f = e^{it(x)}$ . Then

$$\delta(f) = \mathbb{E}(1 - \cos t(x)).$$

Since  $1 - \cos(mt) \leq m^2(1 - \cos t)$ ,

$$\operatorname{Re}(\omega^{-m} \mathbb{E}f^m) = 1 - \mathbb{E}(1 - \cos(mt(x))) \geq 1 - m^2 \delta(f).$$

Therefore  $|\mathbb{E}f^m| \geq 1 - m^2 \delta(f)$ . □

**Lemma 2** (PHNC failure creates a bad cyclic line). Suppose that for some  $A \neq 0$ ,

$$\#\{q \in \mathcal{Q}_\ell(Q) : A \cdot x_q = 0\} \geq (1 - \varepsilon)N.$$

Then for every  $1 \leq j \leq \ell - 1$ ,

$$\delta_Q(jA) \leq 2j^2 \varepsilon.$$

In particular, if  $\varepsilon \leq (4\ell^2)^{-1}$ , then

$$\{A, 2A, \dots, (\ell - 1)A\} \subseteq B_{1/2}(\ell, Q),$$

and hence

$$|B_{1/2}(\ell, Q)| \geq \ell - 1.$$

*Proof.* If  $A \cdot x_q = 0$  on a set of mass at least  $1 - \varepsilon$ , then  $\chi_A(q) = 1$  on that set and  $|\chi_A(q)| = 1$  everywhere. Hence

$$|M_Q(A)| \geq (1 - \varepsilon) - \varepsilon = 1 - 2\varepsilon.$$

Apply [Lemma 1](#) to  $f = \chi_A$ . Since  $\chi_A^j = \chi_{jA}$ , the result follows. □

**Hypothesis 1** (Kummer-sparsity). For a given  $(\ell, Q)$ ,

$$|B_{1/2}(\ell, Q)| < \ell - 1.$$

**Theorem 1** (Kummer-sparsity implies PHNC at that scale). If [Hypothesis 1](#) holds, then for every nonzero  $A \in \mathbb{F}_\ell^2$ ,

$$\#\{q \in \mathcal{Q}_\ell(Q) : A \cdot x_q = 0\} < \left(1 - \frac{1}{4\ell^2}\right) N.$$

*Proof.* This is the contrapositive of [Lemma 2](#). □

**Remark 1.** [Theorem 1](#) is intentionally modest in exponent. The point is not optimization. The point is that PHNC failure is not isolated: it forces an entire projective line of medium-deficit characters.

## 4 The exact mean-square route

Let

$$\mu(v) = \frac{1}{N} \#\{q \in \mathcal{Q}_\ell(Q) : x_q = v\}, \quad v \in \mathbb{F}_\ell^2.$$

Then

$$M_Q(A) = \sum_{v \in \mathbb{F}_\ell^2} \mu(v) e^{2\pi i A \cdot v / \ell}.$$

**Theorem 2** (Bad-character mean square). *One has the exact identity*

$$\sum_{A \in \mathbb{F}_\ell^2 \setminus \{0\}} |M_Q(A)|^2 = \ell^2 \sum_{v \in \mathbb{F}_\ell^2} \mu(v)^2 - 1.$$

*Proof.* Finite Plancherel on  $\mathbb{F}_\ell^2$  gives

$$\sum_{A \in \mathbb{F}_\ell^2} |M_Q(A)|^2 = \ell^2 \sum_v \mu(v)^2.$$

The  $A = 0$  term is 1. □

**Corollary 1** (Mean-square sparsity criterion). *If*

$$\ell^2 \sum_v \mu(v)^2 - 1 < \frac{\ell - 1}{4},$$

*then Hypothesis 1 holds and therefore Theorem 1 applies.*

*Proof.* Every  $A \in B_{1/2}$  has  $|M_Q(A)| \geq 1/2$ . Thus

$$\frac{1}{4} |B_{1/2}(\ell, Q)| \leq \sum_{A \neq 0} |M_Q(A)|^2.$$

Apply Theorem 2. □

## 5 Why the route is a barrier, not a proof

Corollary 1 is exact and useful, but it is not free. It asks for deterministic vector-fiber equidistribution of the Kummer log vectors  $x_q$ . This is a stronger distributional object than PHNC itself.

**Warning 1** (Deterministic vector-fiber barrier). *The mean-square route proves PHNC only after one proves a collision-energy bound for the deterministic prime measure  $\mu$ . Classical finite orthogonality does not supply that bound; it merely names it:*

$$\sum_v \mu(v)^2.$$

**Proposition 1** (Affine hyperplane trap). *Let  $A_0 \neq 0$  and  $t \in \mathbb{F}_\ell^*$ . Let  $\mu$  be the uniform probability measure on the affine line*

$$\{v \in \mathbb{F}_\ell^2 : A_0 \cdot v = t\}.$$

*Then for every nonzero  $A$ , the identity hyperplane mass  $\mu\{v : A \cdot v = 0\}$  is at most  $1/\ell$ , except for  $A \in \mathbb{F}_\ell^* A_0$ , where it is 0. Thus linear identity non-concentration holds strongly. But*

$$|M(A_0)| = 1.$$

*Proof.* If  $A$  is not a scalar multiple of  $A_0$ , the two affine/linear equations cut a single point inside the affine line, so the mass is  $1/\ell$ . If  $A = jA_0$ , then  $A \cdot v = jt \neq 0$  throughout the support. On the other hand,

$$M(A_0) = \sum_v \mu(v) e^{2\pi i A_0 \cdot v / \ell} = e^{2\pi i t / \ell}.$$

□

**Remark 2.** *This is the same trap exposed from a different angle in the Bernoulli-Kummer smoothing paper. Linear PHNC controls mass at identity hyperplanes. Deterministic Fourier smoothing and bad-character mean-square bounds also see affine hyperplane concentration.*

## 6 Analytic status

The structural result of this note can be summarized as:

bad-character sparsity  $|B_{1/2}| < \ell - 1 \implies \text{PHNC at scale } (\ell, Q).$

The missing theorem is the sparsity bound.

**Problem 1** (Fixed-radical Kummer bad-character sparsity). *Prove, uniformly for  $\ell \leq (\log Q)^B$ , that*

$$|B_{1/2}(\ell, Q)| < \ell - 1.$$

**Warning 2** (No BFI plug-in). *Bombieri–Friedlander–Iwaniec type theorems concern primes in arithmetic progressions with well-factorable weights. They do not directly bound the number of fixed-radical Kummer characters whose prime averages are nearly unimodular.*

**Warning 3** (No generic large-sieve plug-in). *Known large sieves for Hecke or  $n$ -th order character families either average over different families than the fixed radicals 2, 3, keep  $n$  fixed, or carry conductor and degree dependence that is not sufficient for the high- $\ell$  seam. A valid plug-in would need to be a fixed-radical, growing- $\ell$  Kummer large sieve or zero-density theorem with explicit dependence strong enough to force  $|B_{1/2}| = o(\ell)$ .*

**Remark 3** (Chebotarev range caution). *For each fixed  $\ell$ , Chebotarev gives eventual Kummer equidistribution. A growing- $\ell$  range must be read from a precise effective Chebotarev theorem and the discriminant/conductor of*

$$\mathbb{Q}(\zeta_\ell, (2^r 3^s)^{1/\ell}).$$

*This note does not assert a paper-grade exponent for that range. It only records that the high seam*

$$(\log Q)^{1-\varepsilon} < \ell \leq (\log Q)^B$$

*is not reached by the bad-character argument without a new analytic input.*

## 7 Program consequence

The bad-character route is valuable because it hardens the target:

$$\text{PHNC failure} \implies \text{an entire cyclic line of bad characters.}$$

That is real amplification. But the route does not bypass the need for high- $\ell$  distribution of fixed-radical Kummer characters. It should therefore be cited as a criterion/barrier note, not as a proof of PHNC.

The next analytic target is either:

1. a fixed-radical Kummer large-sieve or zero-density theorem proving [Problem 1](#); or
2. a Bernoulli-weighted extremal theorem showing that the actual random obstruction law avoids the deterministic affine trap under weaker weighted non-concentration.

## 8 Conclusion

V12 strengthens the program by removing a false hope and replacing it with a theorem-grade criterion. A single PHNC failure is expensive: it creates  $\ell - 1$  medium-bad characters. To turn that expense into a contradiction, one needs a genuine high- $\ell$  fixed-radical Kummer sparsity theorem. That theorem is the wall.

## References

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